

OCOM5100M – Programming for Data Science

Assessment 2 – Marking Criteria

Data Analysis and Modelling Project

Part 1 – Data Analysis Project

Criterion	Not Met 0 - 49	Basic 50-59	Good 60-69	Excellent 70 - 100
1.1 Project Aim and Dataset Description	The aim is absent, unclear or unsuitable. Two measurable objectives are not defined. The dataset source is missing or unreliable, and its contents, quality or suitability are described inaccurately or only superficially.	A relevant aim and two objectives are provided, although one or both may be broad or only partly measurable. The source and main dataset contents are identified. Basic comments are made on quality and suitability, with limited critical detail.	The aim is clear and well motivated. Two specific, measurable and achievable objectives are closely linked to the dataset and domain. The source, contents and structure are described accurately, with good discussion of quality, reliability, limitations and suitability.	The aim is precise, compelling and strongly grounded in the domain. Both objectives are insightful, measurable and clearly drive the analysis. Dataset provenance, contents and structure are explained comprehensively, with a critical and well-evidenced evaluation of quality, reliability, limitations and fitness for purpose.
1.2 Exploratory Data Analysis and Code	The analysis is incomplete, incorrect or does not run reliably. Code is poorly structured, uses inappropriate methods or libraries, or contains substantial irrelevant work. Naming, comments and output interpretation are absent or unclear; outputs may be missing.	The notebook performs a basic, mostly correct exploration using broadly appropriate tools. Code organisation, naming and comments are adequate but inconsistent. Main outputs are visible and briefly explained, although interpretations are mainly descriptive or some choices are weakly justified.	The workflow is correct, purposeful and easy to follow. Code is well organised, concise and appropriately commented, with descriptive names and suitable use of pandas, core Python and visualisation tools. Outputs are interpreted clearly and linked to subsequent analysis decisions.	The EDA is rigorous, selective and highly coherent. Code is robust, reproducible and efficiently structured, with comments explaining purpose rather than restating syntax. Every output contributes to the analytical story, and interpretations reveal strong critical judgement without unnecessary or prohibited processing.
1.3 Objectives Analysis	One or both objectives are not answered. Visual evidence is absent, unsuitable or unclear. Specific numerical values are not reported accurately, and findings are not interpreted in the context of the domain.	Both objectives are addressed with at least one relevant visualisation each. Some numerical evidence is quoted, though selection or interpretation may be limited. Conclusions are broadly reasonable but remain mainly descriptive.	Each objective is answered through appropriate analysis and clear visual evidence. Specific values are accurately selected and interpreted in domain terms. Findings are logically explained and supported by the code, outputs and visualisations.	The analysis provides a convincing and insightful answer to each objective. Visualisations are highly effective, numerical evidence is precise, and interpretation demonstrates strong domain awareness, critical reasoning and appropriate consideration of uncertainty or alternative explanations.
1.4 Insights and Reflections	The summary is missing, inaccurate or disconnected from the analysis. Main findings, unexpected results or limitations are omitted or asserted without evidence.	A basic summary of the main findings is provided. An unexpected aspect and some limitations are mentioned, although the discussion may be general or weakly connected to the results.	The section synthesises the most important findings accurately, identifies a meaningful unexpected result and discusses relevant limitations. Reflection is clearly grounded in the analysis and avoids overclaiming.	The section offers a concise, critical synthesis that prioritises the most important insights. The unexpected finding is thoughtfully interpreted, and limitations and implications are evaluated with nuance, specificity and clear connection to the evidence.

Part 2 – Machine Learning Pipeline

Criterion	Not Met 0-49	Basic 50-59	Good 60-69	Excellent 70-100
2.1 Preprocessing and Pipelines	Preprocessing is incomplete, inappropriate or introduces data leakage. Numerical and categorical features are mishandled, the test set is used during development, or a valid scikit-learn pipeline is not produced. Decisions are not explained.	A functional preprocessing workflow and pipeline are produced with broadly appropriate handling of the main feature types. Major leakage is avoided. The reflection gives a basic rationale, although some choices are generic, inconsistent or insufficiently linked to the dataset.	A coherent, leakage-safe preprocessing pipeline is correctly implemented. Choices such as imputation, encoding, scaling or feature treatment are appropriate to the data and models. The reflection clearly justifies decisions and recognises relevant alternatives or limitations.	The pipeline is robust, reproducible and carefully engineered for the dataset. Preprocessing decisions are critically justified with strong awareness of leakage, model requirements and trade-offs. Alternatives are evaluated proportionately, and the resulting workflow integrates cleanly with all later modelling stages.
2.2 Baseline Model Training	Two valid baseline classifiers are not trained, or models are fitted outside complete pipelines. Code is incorrect or inconsistent with the preprocessor, and classifier choices are not justified.	Two working classification pipelines are trained using the shared preprocessor. The classifiers provide usable baselines, and a brief but reasonable justification is given.	Two appropriate and meaningfully contrasting baseline models are implemented correctly as complete pipelines. The rationale is clear, dataset-aware and connected to expected model behaviour.	The baseline choices are strategically selected to establish an informative comparison. Both pipelines are clean, reproducible and consistent, and the concise rationale demonstrates strong understanding of assumptions, strengths and likely trade-offs.
2.3 Cross-Validation	Cross-validation is absent, incorrectly applied or leaks test information. Results do not include an appropriate metric beyond accuracy, or mean and standard deviation are missing. Model selection is unsupported.	Both baseline pipelines are evaluated using a broadly appropriate cross-validation procedure. At least one additional metric and the mean and standard deviation are reported. The comparison and model choice are present but may be mainly descriptive.	Cross-validation is correctly and consistently applied using suitable metrics. Results are clearly reported and interpreted in terms of performance and stability. The selected model is justified using the evidence and the needs of the classification task.	The evaluation is rigorous and thoughtfully interpreted. Metric choice, fold variability and relevant class trade-offs are considered together. The model selected for tuning follows convincingly from the evidence, with uncertainty and practical significance handled appropriately.
2.4 Hyperparameter Tuning	A valid hyperparameter search is not completed, the held-out test set is used for tuning, or the search is not applied to the full pipeline. The best configuration and comparison with baseline performance are missing or incorrect.	One model is tuned using a recognisable search procedure. The best configuration is reported and compared with baseline cross-validation performance. The explanation is broadly reasonable but may provide limited justification or interpretation.	The search strategy, parameter space and scoring choice are appropriate and applied correctly to the full pipeline. The best result is reported clearly, and the reflection evaluates whether the improvement is meaningful, not merely whether it is numerically higher.	Tuning is efficient, leakage-safe and carefully justified. The search space reflects informed model knowledge, and the comparison with baseline performance is nuanced. The discussion recognises uncertainty, diminishing returns and possible overfitting, and avoids overstating small gains.
2.5 Final Model Evaluation and Justification	The final model is not evaluated correctly on a held-out test set. Required metrics or a labelled confusion matrix are missing. Results are misinterpreted, not compared with cross-validation, or no fit-for-purpose judgement is provided.	The final pipeline is evaluated on the test set using precision, recall, F1 and a labelled confusion matrix. A basic comparison with cross-validation is made, and a reasonable recommendation is provided, although	Test performance is accurately interpreted using appropriate metrics and class-level errors. The comparison with cross-validation supports a clear discussion of generalisation. The fit-for-purpose recommendation is balanced, evidence-based and includes	The final evaluation offers a critical synthesis of test results, cross-validation and task requirements. Error trade-offs and limitations are analysed with precision. The recommendation is proportionate, domain-aware and transparent

		interpretation may be limited.	relevant areas for improvement.	about uncertainty, risks and realistic next steps.
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Process Log and Report Quality

Criterion	Not Met 0-49	Basic 50-59	Good 60-69	Excellent 70-100
Process Log	The log is missing, substantially incomplete or exceeds the stated limit without justification. It does not address two specific dead ends, a surprising finding and one evidence-based change. Reflection is generic or disconnected from the submitted work.	All three required areas are addressed: at least two dead ends, one surprising finding and one change for a future attempt. Examples are recognisable and partly linked to outputs, but explanation of causes, learning or consequences may be brief.	The log gives a specific and reflective account of unsuccessful attempts across the work, explains what was learned, identifies a surprising result using a concrete value or output, and proposes a feasible improvement grounded in the submitted analysis and modelling.	The log demonstrates highly effective reflective practice and iterative problem-solving across Parts 1 and 2. Dead ends are analysed rather than merely listed, the surprising finding is interpreted precisely, and the proposed change is focused, feasible and convincingly derived from the evidence.
Report Quality	Placeholder text remains, required headings are changed, outputs are missing, or the notebook contains unresolved errors. Writing and organisation make the work difficult to follow, and code, outputs and commentary are poorly connected.	The notebook is complete and generally readable, with headings retained and most outputs visible. Writing is understandable but may contain minor errors, repetition or uneven organisation. Connections between code, outputs and commentary are adequate.	The notebook is polished, coherent and professionally organised. All outputs are visible, instructional text is removed, writing is clear and concise, and the relationships between code, results and interpretation are easy to follow.	The submission is exceptionally coherent and professional. Presentation is consistent, concise and virtually error-free; section structure is preserved; outputs are complete; and code, visualisations and commentary form a seamless analytical narrative within the stated limits.